ICLAB 2017Fall Final Exam

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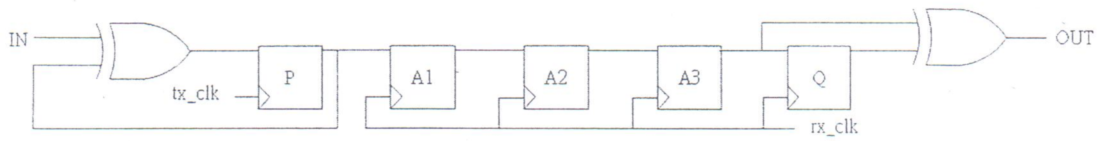
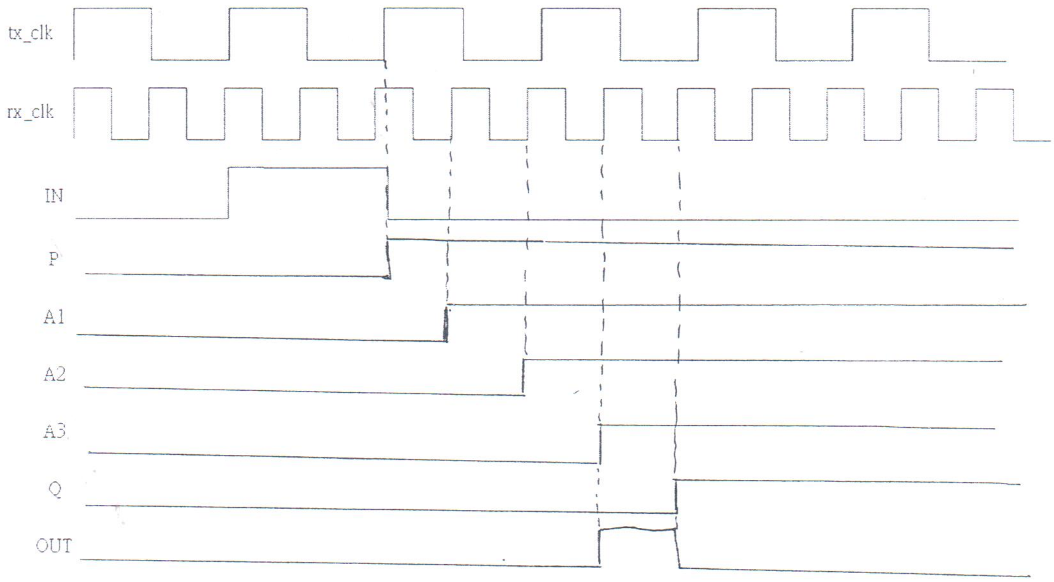
1. [10%] Please explain STA (static timing analysis) and DTA (dynamic timing analysis) briefly and compare them.

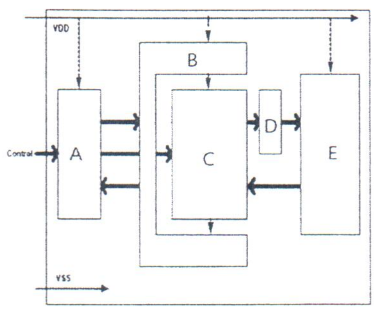
2. [4%] (a) Please explain why we need a synchronizer in our design and its functionality.
   [4%] (b) What may happened if we don't use XOR gates in a synchronizer?
   [6%] (c) Complete the waveform below





3. [6%] When using clock gating, in turns of power consumption, which gating is better? AND-Gating or OR-Gating? Please justify your answer.

4. [8%] Down below is the figure of power gating.
   A is "Power gating controller"
   B is "Power switching fabric"
   C is "Power gated block"
   E is "Always on block"
   What is block "D"?
   Please also briefly describe how to implement the block D using one gate as an example.



5.

A) [6%] Why is Typedef used in systemVerilog?

B) [6%] Write a representation of hBytes if "reg" is used as a datatype instead of hexOctet

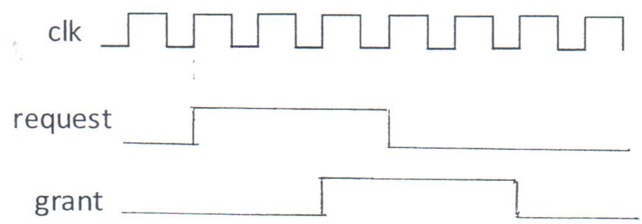
```
typedef reg [7:0] octet;

typedef octet [15:0] hexOctet;

hexOctet hBytes [3:7];
```

6. [6%] What values are constrained by "Limit"? (List them out)

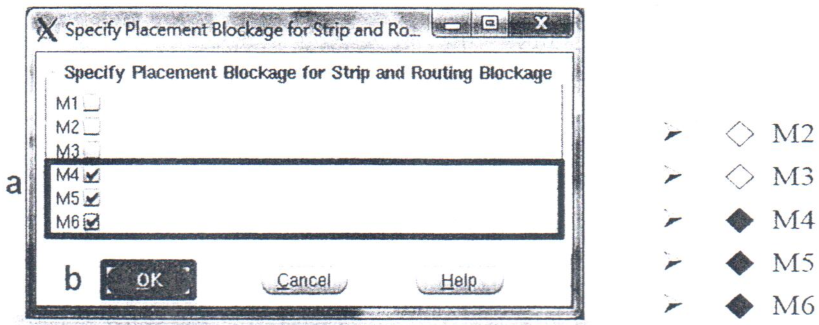
```
constraint Limit{

sa inside { [4:8], 11, 21};

payload.size() <= 4;

}
```

7. [6%] How many state bins are created:
    a. bins s0 = {[0:4]}
    b. bins s1 [] = {[3:9]}
8. [6%] Fill in the waveform below given the property:

```
property p_request_grant;

        @(posedge clock) request ##2 grant ##1 !request ##2 !grant

endproperty
```



9. [4%] What is purpose of the "uniquify" command in the design flow?
10. [6%] What is the cause, and proper solution of IR drop?
11. [4%] In APR flow there is a step before we place the standard cell. **Descript its intention.**
    The step: **Place -> Specify -> Placement Blockage**



12. [8%] What is **Antenna Effect**? List 2 repair method for Antenna Effect.
13. [4%] Please simply describe why **RC extraction** is needed.
14. [6%] Describe the powered simulation methods **vector-based** and **vector-independent**.

DTA: 針对 input vector 來分析 timing,

当 input vector 变多, 时間就会愈久.

2、

a. 当有 block 处於不同 clock domain 时,

就需要 synchronizer 來確保下一個 block 可以接收到 上一個block 完整的資訊.

b. 如果沒加 XOR, 則有可能会收不到上一個 block 的東西, 或是收到的 data 是錯的.

因為 2 個 block 是処於不同的 clock domain.

而 XOR 是確保上一個 block 的 pulse 一定能夠被下一個 block 吃到.

如果沒加 XOR, 下一個 block 有可能就会吃不到上一個 block 給的 invalid!

c.

3. OR-gate.

+6

OR-gate: the gated clock is tied high when registers are turned off.

∴ the first latch circuit will not toggle while data is toggling..

AND-gate: the gated clock is tied low when registers are turn off.

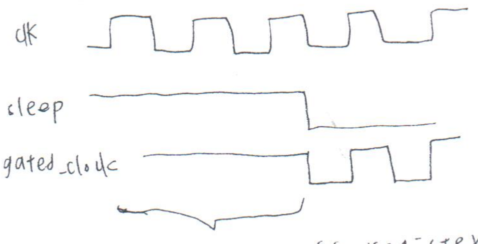
∴ the first latch circuit will toggle while data is toggling.

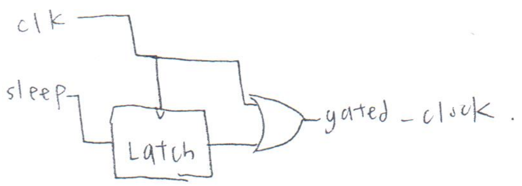
So this one consumes more power.

4.

+6 D: isolation.

Ex. OR-gate





5.

a. 可以 define 自己想要的 type，並且用這個 type 來宣告其他變數.

Ex. typedef enum logic[1:0] {none=2'b00, first=2'b01, second=2'b10} num;

define 一個叫做 "num" 的 data type. +6

b.

hexOctex → reg [7:0] [15:0] hexOctex.

+6

hBytes: reg [7:0][15:0] hBytes [3:7] #

6、

sa、和 payload的size. ↗ ≤4.

↓

只能有1,2,4~8.

+6

7.

8. +6

a. 1 state bins.

b. 7 state bins.

+6

8.

9. 把每個module都当做是一個獨立的block!

+4

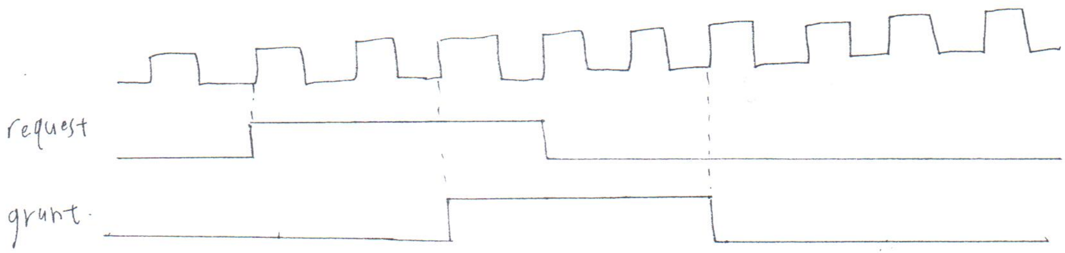
避免和其他module混合而合成錯誤.

10.

+3 solution: add strip (stripes)

cause: 因為电容的充放电,可能会使得电压下降

8.



11、

+3

把 strip 和 routing 和 standard cell 隔開

避免 wire 間的效應影響电路.

12、

+6

metal 和 metal 間会有 oxide，因此形成电位差.

当电位差太大时,就会擊穿 oxide，对电路產生影响.

solution: add oxide, add jumper, add buffer.

13.

+2

R.C. 充放电时会使得电压產生變化.

若电压变化太大就会使得电路無法正確執行.

∴可藉由 RC extration 来分析电压.

14.

+3

vector-based: 藉由 input vector 来分析 power.

vector-independent: 分析电路整体的 power，和 input vector 無関.

辛苦啦! ☺